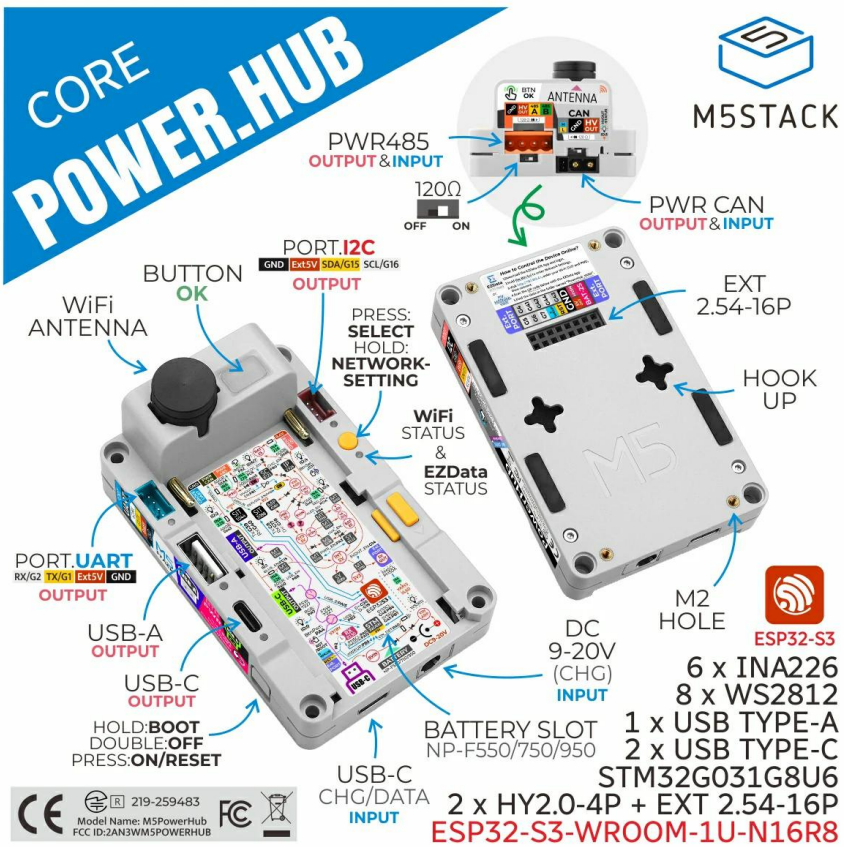
POWERHUB PMIC(STM32)
USER MANUAL



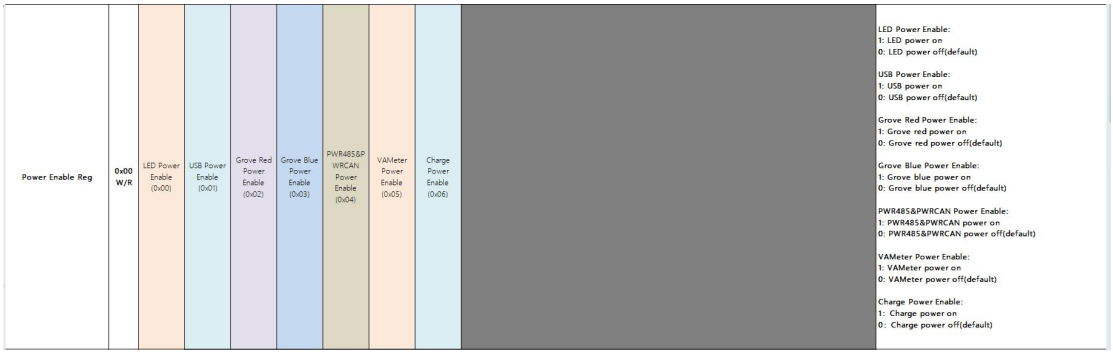
DATE	CONTENT	VERSION
2025-02-27	Firmware update	2.0
2025-10-16	Adds input detection.	2.1

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I、Power Enable Reg

Address	0x00 - 0x06
Access	W/R
Function	Enables or disables the LED power, USB power, Grove Red port power, Grove Blue port power, PWR485 & PWRCAN power, voltage/current meter power, and charging power.



1. LED Power Enable

Controls the power enable of the RGB LED. Write 1 to turn on, write 0 to turn off. The default state is off.

2. USB Power Enable

Controls the power enable of the USB port. Write 1 to turn on, write 0 to turn off. The default state is off.

3. Grove Red Power Enable

Controls the power enable of the Grove Red port. Write 1 to turn on, write 0 to turn off. The default state is off.

4. Grove Blue Power Enable

Controls the power enable of the Grove Blue port. Write 1 to turn on, write 0 to turn off. The default state is off.

5. PWR485&PWRCAN Power Enable

Controls the power enable of the PWR485 & PWRCAN port. Write 1 to turn on, write 0 to turn off. The default state is off.

6. VAMeter Power Enable

Controls the power enable of the voltage/current meter. Write 1 to turn on, write 0 to turn off. The default state is off.

7. Charge Power Enable

Controls the charging enable. Write 1 to turn on, write 0 to turn off. The default state is off.

II、USB Mode

Address	0x10
Access	W/R
Function	Controls the USB port mode, switching between device mode and host mode.



0：Slave mode，USB->ESP32(default)
1：Host mode(USB TypeA)，ESP32->USB TypeA device
2：Host mode(USB TypeC)，ESP32->USB TypeC device

III、PWR485&PWRCAN Config

Address	0x20 - 0x24
Access	W/R
Function	Controls the SC8721 voltage, maximum current, power switch (physical switch), and power direction (output or input).



1. Voltage（LSB&MSB）

Configures the output voltage of the SC8721, range 3000 mV to 20000 mV, with a step size of 20 mV.

2. Current Limit Value

Calculation formula: Iout limit (mA) = (Current Limit Value + 1) * 8 / 0.62
Example: To set the current limit to 1000 mA, Current Limit Value = 1000 * 0.62 / 8 - 1 = 76

Note: The total output power must not exceed 20 W. If the output voltage is set high, the output current register will be limited to a lower value.

3. Power Output Control

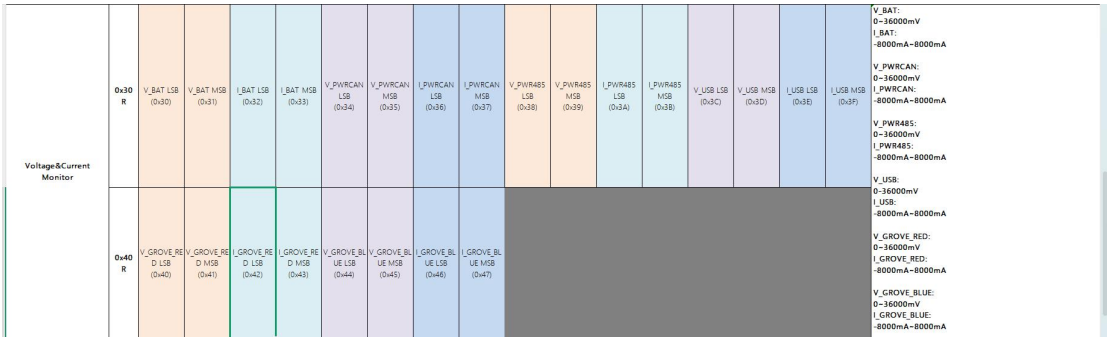
Configures the SC8721 output using a MOSFET for physical switching. Write 1 to turn on the power output, write 0 to turn off. The default state is off. (Note: When the power direction is set to output, it will be automatically turned on.)

4. Power Direction Control

Configures the input/output direction of the SC8721. Write 1 to set the direction to output, write 0 to set the direction to input. The default state is input.

IV、Voltage&Current Monitor

Address	0x30 - 0x47
Access	R
Function	Retrieves the voltage and current from the INA226 on each channel.



V、Detect Status

Address	0x50~0x51
Access	R
Function	Retrieves the charging status and power input status.



1、Charge Status

0：no charge

1：charging

2：error

2、VIN Status

0：no input

1：input present

VI、LED Color

Address	0x60 - 0x7F
Access	W/R
Function	Controls the color of 8 LEDs.



Updates the color of the corresponding LED by writing the corresponding RGB values.

VII、LED Brightness

Address	0x80 - 0x87
Access	W/R
Function	Controls the brightness of 8 LEDs.



Updates the brightness of the corresponding LED by writing the corresponding RGB values.

VIII、Button

Address	0xA0
Access	R
Function	Retrieves the button status.



0:press;
1:no press

IX、Wakeup Source

Address	0xB0 - 0xB2
Access	W/R
Function	Configures the wake-up source.



1. nSTBY_WKUP

Configures whether to enable RTC clock wake-up. Write 0 to disable, write 1 to enable. The default state is enabled.

2. VIN_DET

Configures whether to enable VIN insertion wake-up. Write 0 to disable, write 1 to enable. The default state is enabled.

3. VUSB_DET

Configures whether to enable VUSB insertion wake-up. Write 0 to disable, write 1 to enable. The default state is enabled.

Note: At least one of the three wake-up sources must be enabled. If all three wake-up sources are disabled, sleep will not be allowed.

X、RTC Time

Address	0xC0 - 0xC6
Access	W/R
Function	Configures the RTC clock.



Sec: 0-59
Minute: 0-59
Hour: 0-23
Day: 1-31
Month: 1-12
Year: 0-168
Week Day:0x01, Sunday; 0x02, Monday; 0x04, Tuesday; 0x08, Wednesday;
0x10: Thursday; 0x20: Friday; 0x40: Saturday; 0x80: Sunday

XI、RTC Alarm

Address	0xD0 - 0xD3
Access	W/R
Function	Configures the RTC alarm.



Minute: 0-59
Hour: 0-23
Day: 1-31
Alarm Enable: 1: Alarm Enable; 0: Alarm Disable

XII、Bootloader Version

Address	0xFC
Access	R
Function	Retrieves the bootloader version number.



XIII、Firmware Version

Address	0xFE
Access	R
Function	Retrieves the firmware version number.



XIV、I2C Address

Address	0xFF
Access	W/R
Function	Sets the I2C address.

